Polymer Chain Dynamics & Scaling Analysis

Radius of Gyration (R_g^2) Convergence and End-to-End Vector Relaxation ($C_R(t)$)

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Batch Simulation Setup

- **Polymer Sizes Investigated (N)**:

$$N \in \{10, 20, 30, 40, 50, 100, 150, 200, 250, 300\}$$

- **Equilibration Steps (N_{eq})**: Scaled quadratically with chain length N:

$$N_{\text{eq}} = 500\, N^2$$

- **Total Production Steps (N_{prod})**: Scaled quadratically with chain length N:

$$N_{\text{prod}} = 1500\, N^2$$

- **Chunking Interval**: Trajectory observables are recorded every chunk $= 500$ steps, yielding $M = N_{\text{prod}}/500 = 3N^2$ recorded frames per trajectory.

Quadratic Step Scaling (N^2)

Rouse-like polymer relaxation times scale approximately as $\tau \propto N^{2\nu+1} \approx N^{2.2}$. Scaling $N_{\text{prod}} = 1500\, N^2$ ensures that even the largest chains ($N = 300$, $N_{\text{prod}} = 1.35 \times 10^8$ steps) span thousands of independent relaxation times.

Summary Metrics Extracted

1. Final $\langle \text{ROG} \rangle \equiv \langle R_g^2 \rangle_M$
2. ROG Std Dev $\equiv \sigma_{R_g^2}$
3. Integral Relaxation Time (τ_{int})
4. Exponential Relaxation Time (τ_{exp})
5. Total Production Steps (N_{prod})

Methodology I: Structural Observables (Final ⟨ROG⟩ & Std Dev)

For a polymer of N monomers with instantaneous 3D coordinates $\boldsymbol{r}_i(t_k)$ recorded across M production frames ($k = 0, 1, \ldots, M-1$):

1. Instantaneous Squared Radius of Gyration (`rogs` $= R_g^2(t_k)$)

$$R_g^2(t_k) = \frac{1}{N} \sum_{i=0}^{N-1} \left\| \boldsymbol{r}_i(t_k) - \boldsymbol{r}_{\mathrm{cm}}(t_k) \right\|^2, \qquad \text{where} \quad \boldsymbol{r}_{\mathrm{cm}}(t_k) = \frac{1}{N} \sum_{i=0}^{N-1} \boldsymbol{r}_i(t_k)$$

2. Running Mean (`running_mean`) & Final ⟨ROG⟩

The cumulative running mean up to frame $m \in \{1, \ldots, M\}$ and the terminal ensemble average (`running_mean[-1]`) reported in the summary table are:

$$\langle R_g^2 \rangle_m = \frac{1}{m} \sum_{k=0}^{m-1} R_g^2(t_k) \quad \rightarrow \quad \langle R_g^2 \rangle_M = \frac{1}{M} \sum_{k=0}^{M-1} R_g^2(t_k)$$

3. Standard Deviation of R_g^2 (`ROG Std Dev`)

Computed across all M recorded production frames via `np.std(rogs)`:

$$\textbf{ROG Std Dev} = \sigma_{R_g^2} = \sqrt{\frac{1}{M} \sum_{k=0}^{M-1} \left(R_g^2(t_k) - \langle R_g^2 \rangle_M \right)^2}$$

Let $\boldsymbol{R}(t_k) = \boldsymbol{r}_{N-1}(t_k) - \boldsymbol{r}_0(t_k) \in \mathbb{R}^3$ denote the instantaneous end-to-end vector at frame k, with lag time $t_k = k \cdot \Delta t_{\text{chunk}}$ (in simulation steps).

4. FFT-Based Normalized End-to-End Vector Autocorrelation ($C_R(t_k)$)

$$C_{\text{raw}}(t_k) = \frac{1}{M-k} \sum_{m=0}^{M-1-k} \boldsymbol{R}(t_m) \cdot \boldsymbol{R}(t_{m+k}), \qquad C_R(t_k) = \frac{C_{\text{raw}}(t_k)}{C_{\text{raw}}(0)} = \frac{\langle \boldsymbol{R}(0) \cdot \boldsymbol{R}(t_k) \rangle}{\langle \boldsymbol{R}^2 \rangle}$$

5. Integral Relaxation Time (τ_{int})

Let $k_0 = \min\{k : C_R(t_k) \leq 0\}$ be the first zero-crossing index ($k_0 = M$ if none). Integrated via the composite trapezoidal rule (`np.trapezoid`):

$$\tau_{\text{int}} = \int_0^{t_{k_0-1}} C_R(t)\, dt \approx \sum_{k=0}^{k_0-2} \frac{C_R(t_k) + C_R(t_{k+1})}{2} \Delta t_{\text{chunk}}$$

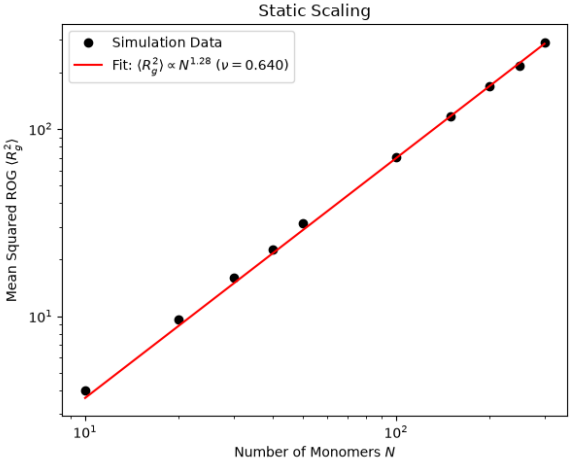
6. Exponential Relaxation Time (τ_{exp})

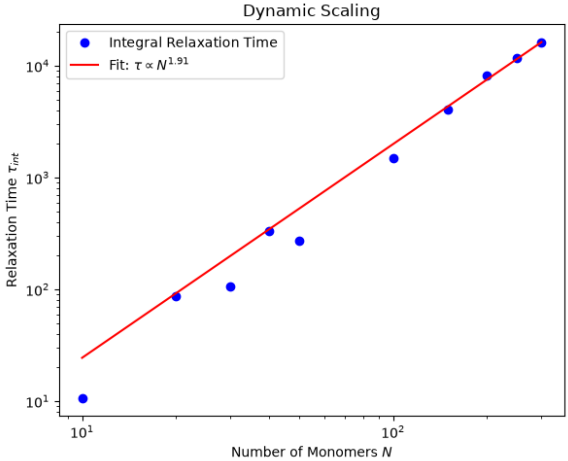
Assuming $C_R(t) \sim A \exp(-t/\tau_{\text{exp}})$, a linear least-squares fit (`np.polyfit`) of $\ln C_R(t_k)$ vs. t_k is performed over the pre-cutoff window $0.15 \leq C_R(t_k) \leq 0.85$:

$$\ln C_R(t_k) = s \cdot t_k + b \implies \tau_{\text{exp}} = -\frac{1}{s}$$

(Fallback if < 3 points in range: first $1/e$ crossing $C_R(t_k) \leq e^{-1}$).

Curve Fit of Simulation Data





Quantitative Summary Table Across Polymer Sizes (N)

N (Particles)	Final $\langle$ROG$\rangle$ ($\langle R_g^2 \rangle$)	ROG Std Dev ($\sigma_{R_g^2}$)	τ_{int} (steps)	τ_{exp} (steps)	Total Prod Steps
10	4.0283	1.2802	10.6999	10.6916	150,000
20	9.5520	3.2684	87.7221	132.5621	600,000
30	15.9823	5.5808	107.4081	127.2257	1,350,000
40	22.5506	8.2170	330.5696	473.6647	2,400,000
50	31.1611	11.4339	274.3272	360.3458	3,750,000
100	70.9265	28.2231	1,496.5783	1,528.1411	15,000,000
150	117.0141	42.8667	4,052.1894	5,242.8174	33,750,000
200	169.5461	60.3666	8,141.9466	10,259.2156	60,000,000
250	216.8072	85.7008	11,634.7945	16,073.8872	93,750,000
300	290.0355	118.5301	16,045.1805	20,882.9344	135,000,000
Mean	**94.7604**	**36.5468**	**4,218.1416**	**5,509.1485**	**34,575,000**
Median	**51.0438**	**19.8285**	**913.5740**	**1,000.9029**	**9,375,000**
Std	**100.3560**	**40.2179**	**5,773.9283**	**7,647.9595**	**47,194,454**

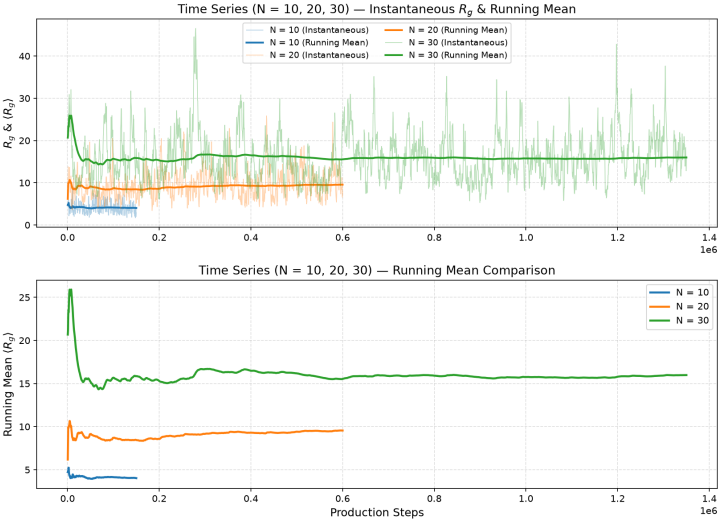
Power-Law Scaling Exponents ($N \in [10, 300]$)

- **Flory Scaling**: $\langle R_g^2 \rangle \propto N^{2\nu}$ gives $2\nu = 1.2468 \implies \boldsymbol{\nu = 0.6234}$ (3D self-avoiding walk theory: $\nu \approx 0.588$).
- **Relaxation Scaling**: $\tau_{\text{int}} \propto N^{2.1070}$ and $\tau_{\text{exp}} \propto N^{2.1319}$ (consistent with Rouse scaling $z = 2\nu + 1 \approx 2.2$).

Notable Trends & Outliers

- Minor statistical non-monotonicity in $\tau_{\text{int}}, \tau_{\text{exp}}$ between $N = 40$ (330.57) and $N = 50$ (274.33).
- Relative fluctuation ratio $\sigma_{R_g^2}/\langle R_g^2 \rangle$ remains invariant around ≈ 0.32–0.41 across all N.

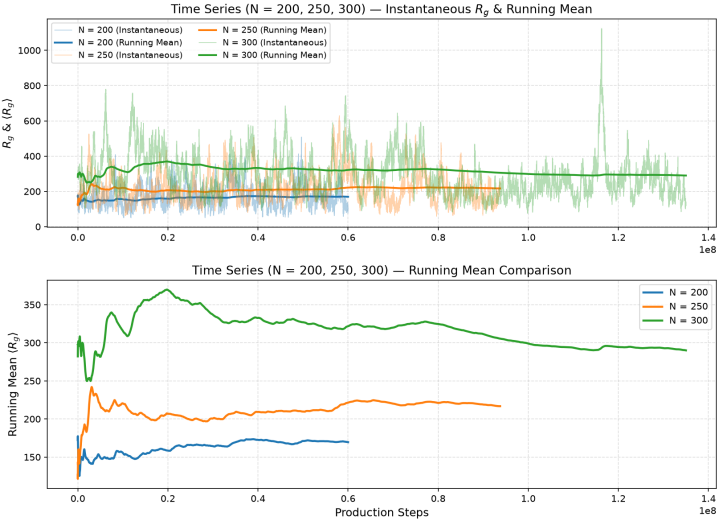
Small-N Trajectory Convergence ($N = 10, 20, 30$)



Key Observations ($N = 10, 20, 30$)

- **Rapid Equilibration**: Short relaxation times ($\tau_{\text{int}} \in [10.70, 107.41]$ steps) allow the running mean $\langle R_g^2 \rangle$ to plateau early in the production run.
- **Plateaus**:
  - $N = 10$: $\langle R_g^2 \rangle = 4.03 \pm 1.28$
  - $N = 20$: $\langle R_g^2 \rangle = 9.55 \pm 3.27$
  - $N = 30$: $\langle R_g^2 \rangle = 15.98 \pm 5.58$

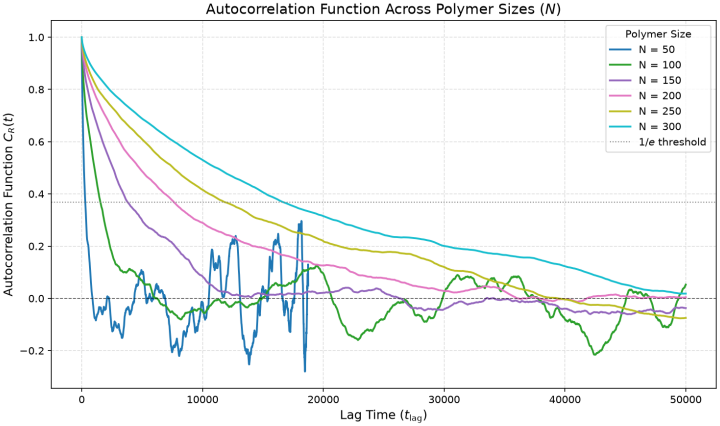
Large-N Trajectory Convergence ($N = 200, 250, 300$)



Key Observations ($N = 200, 250, 300$)

- **Large Conformational to Reach End Equilibration**: Instantaneous R_g^2 exhibits broad fluctuations ($\sigma_{R_g^2} = 60.37, 85.70, 118.53$), reflecting slow global chain reconfiguration.
- **Running Mean Stabilization**:
  - $N = 200$: $\langle R_g^2 \rangle = 169.55$
  - $N = 250$: $\langle R_g^2 \rangle = 216.81$
  - $N = 300$: $\langle R_g^2 \rangle = 290.04$

End-to-End Vector Autocorrelation Decay ($C_R(t_{\text{lag}})$)



Relaxation Spectrum Analysis

- **Systematic Slowing Down**: $C_R(t)$ decay curves broaden monotonically with chain length N for $N \geq 100$.
- **Zero-Crossing Cutoff**: Integrating up to the first $C_R(t) \leq 0$ crossing suppresses long-time tail noise.